



2022 YEAR 11 ASSESSMENT TASK 2

Mathematics Extension

General Instructions

- Reading time – 5 minutes
- Working time – 1 hour 30 minutes
- Write using black pen.
- Calculators approved by NESA may be used.
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations.
- Answer all questions in the booklet provided.
- A reference sheet is provided at the back of this paper

Total marks: 59

Section I – 10 marks (pages 2 – 4)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II – 49 marks (pages 5 – 8)

- Attempt Questions 11 – 14
- Allow about 1 hour and 15 minutes for this section.

Section I

10 marks

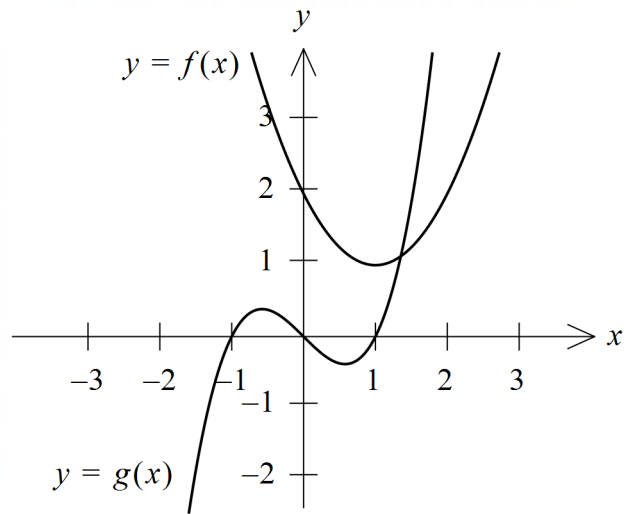
Attempt questions 1 - 10

Allow about 15 minutes for this section

Use the multiple choice answer page in your booklet to answer Questions 1 – 10.

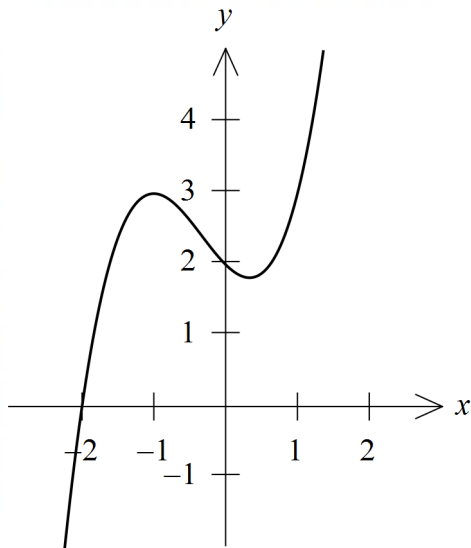
- 1) What is the solution of $|x - 5| \leq 2$?
- A) $-7 \leq x \leq -3$ B) $x \leq -7, x \geq 3$
- C) $3 \leq x \leq 7$ D) $x \leq 3, x \geq 7$
- 2) The function $f(x)$ is defined as $f(x) = x^3 - 4$. Which expression is equal to $f^{-1}(x)$?
- A) $\frac{1}{x^3} - 4$ B) $\frac{1}{x^3 - 4}$ C) $\sqrt[3]{x+4}$ D) $\sqrt[3]{x} + 4$
- 3) If there are 5 boxes to store 22 pens, what does the pigeonhole principle state?
- A) At least 4 boxes will contain more than 4 pens.
- B) At least 1 box will contain more than 4 pens.
- C) At most 4 boxes will contain more than 4 pens.
- D) At least 1 box will contain less than 4 pens.
- 4) How many three-digit PIN codes can be formed from the digits 0, 1, 2, 3, 4 and 5 if repetition is not allowed?
- A) 720 B) 24 C) 120 D) 60
- 5) The polynomial $P(x) = 2x^3 + ax - 3$ has a remainder of -1 when it is divided by $(x+1)$. What will be the remainder if it is divided by $(x-1)$?
- A) -6 B) -5 C) 1 D) 5
- 6) The polynomial $P(x) = x^3 - x^2 - 5x - 3$ has a double root at $x = \alpha$. What is the value of α ?
- A) $-\frac{5}{3}$ B) -1 C) 1 D) $\frac{5}{3}$

- 7) The graphs of a parabolic function $y = f(x)$ and a cubic function $y = g(x)$ are shown below:

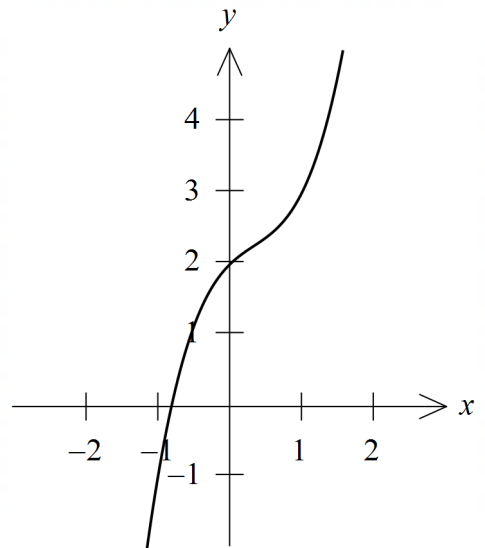


Which of the following represents the graph of $y = f(x) + g(x)$?

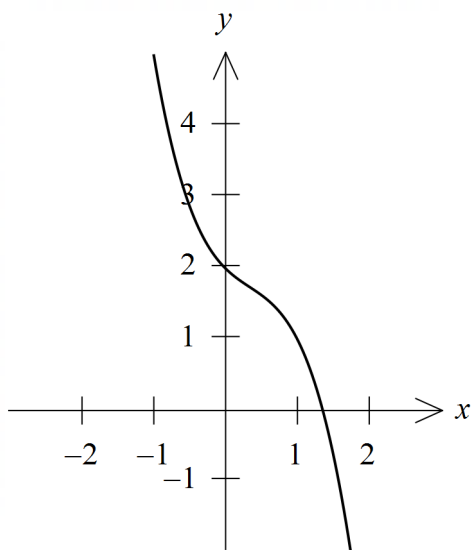
A)



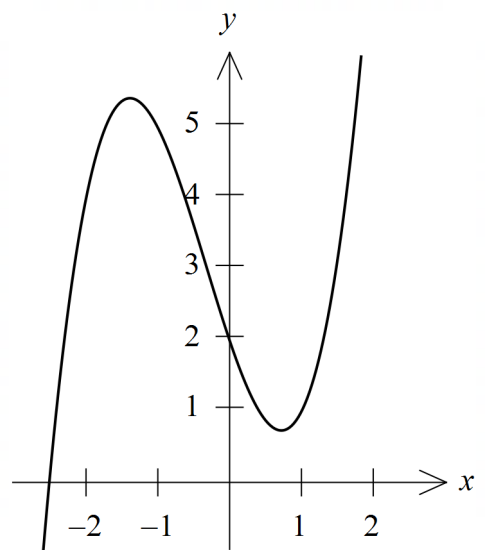
B)



C)



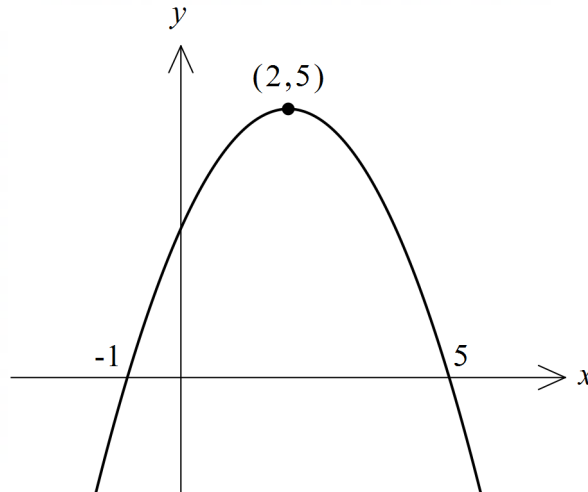
D)



- 8) In how many ways can the letters of the word **INDEPENDENCE** be arranged in a row, if all the letters 'E' being together?

A) $\frac{12!}{4!3!2!}$ B) $\frac{9!}{4!3!2!}$ C) $\frac{9!}{3!2!}$ D) $\frac{1}{55}$

- 9) The graph of $y = f(x)$ is given below:



What is the range of $y = \frac{1}{f(x)}$?

A) $(-\infty, 0) \cup \left[\frac{1}{5}, \infty\right)$ B) $\left(0, \frac{1}{5}\right)$
 C) $(-\infty, 0) \cup \left(\frac{1}{5}, \infty\right)$ D) $\left(0, \frac{1}{5}\right]$

- 10) Given $f(x)$ is a quadratic function whose graph is symmetrical about $x = c$, what is the value of $f^{-1}(f(a))$ if $a > c$ and a is NOT in the restricted domain of $f(x)$ such that $f^{-1}(x)$ exists?

A) $a - 2c$ B) $2c - a$ C) $2a - c$ D) a

End of Section I

Section II

49 marks

Attempt Questions 11 - 14

Allow about 1 hour 15 minutes for this section

Answer each question on the appropriate pages of your answer booklet. Extra paper is available.
Your responses should include relevant calculations and/or mathematical reasoning.

Question 11 (14 marks)

- a) Solve for x , and graph the solution on a number line: 3

$$\frac{2}{x-1} > 3$$

- b) A team of five is to be chosen at random from 7 women and 5 men.

- i) In how many ways can this team be chosen? 1

- ii) What is the probability that the team will consist of at most 2 women? 2

- c) i) Find the inverse function of $f(x) = \frac{3}{x-2} + 1$. 2

- ii) Hence show that $f^{-1}(f(x)) = x$. 1

- d) The polynomial $P(x) = 2x^3 - 10x^2 - 40x + 175$ has roots α , β , and γ . Find the value of:

- i) $\alpha + \beta + \gamma$ 1

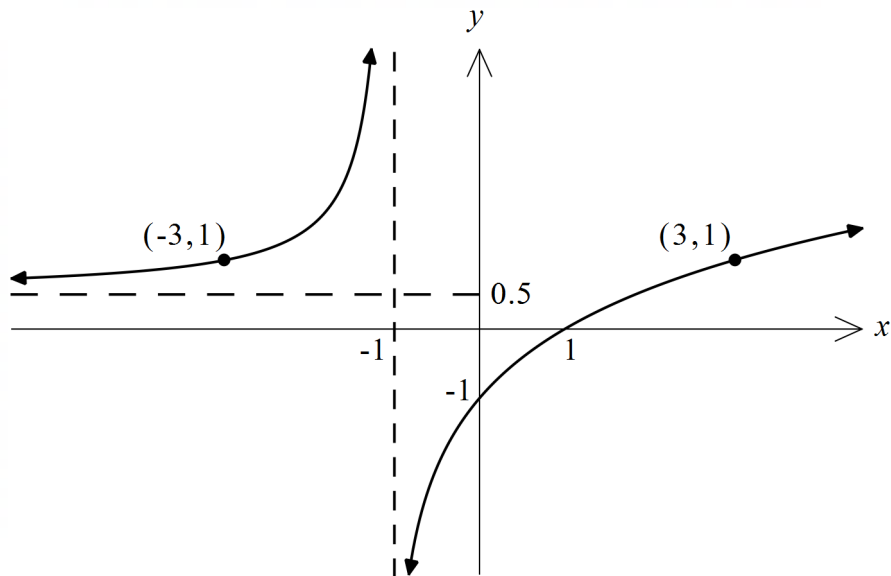
- ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 2

- iii) $\alpha^2 + \beta^2 + \gamma^2$ 2

End of Question 11

Question 12 (12 marks)

- a) The diagram shows the graph of $y = f(x)$.



On separate axes provided in the answer booklet, sketch the graphs of each of the following functions, clearly labelling all important features on your graph.

- i) $y = f(|x|)$ 1
- ii) $y = \sqrt{f(x)}$ 2
- iii) $y = \frac{1}{f(x)}$ 2
- b) A bowl contains 10 red cubes and 10 blue cubes, all of the same size. A woman reaches into the bowl and selects cubes at random without looking at them.
- i) What is the minimum number of cubes she must select to be sure of having at least three cubes of the same colour? 1
- ii) What is the minimum number of cubes she must select to be sure of having at least three blue cubes? 1
- c) Solve $\frac{1}{x(x+2)} \leq -1$. 3
- d) When $P(x)$ is divided by $(x-3)$, the remainder is 4, and when it is divided by $(x+5)$, the remainder is 12. What would the remainder be if $P(x)$ is divided by $(x-3)(x+5)$? 2

End of Question 12

Question 13 (12 marks)

- a) The letters of the word **BEGINNING** are to be arranged in a row.
- i) In how many arrangements are the letter G's together? 1
 - ii) How many arrangements will **NOT** contain adjacent N's? 2
(For example, **NGENINBIG** is one such arrangement but the arrangements **NNNBEGHIG** and **GNNNEGHIIN** are NOT.)
- b) Consider the function $f(x) = x^2 + 4x + 1$.
- i) Briefly explain why the domain has to be restricted for an inverse function to exist. 1
 - ii) What is the largest domain containing the value $x = 0$, for which the function has an inverse function $f^{-1}(x)$? 1
 - iii) Show that the inverse function in its restricted domain is $f^{-1}(x) = -2 + \sqrt{x+3}$. 2
 - iv) Sketch the graphs of $y = f(x)$, $y = f^{-1}(x)$ and $y = x$ on the same set of axes within the restricted domain. (DO NOT find any points of intersection) 2
- c) Two roots of the equation $x^3 + px^2 - qx + 3 = 0$ are reciprocals of each other.
- i) Evaluate the third root. 1
 - ii) Hence find the relation between p and q , expressing p in terms of q . 2

End of Question 13

Question 14 (11 marks)

- a) It is given that the polynomial $P(x) = 4x^3 - 13x^2 + 4x + 12$ has a double root at $x = \alpha$.
- i) By differentiating $P(x)$, find the value of α . 2
- ii) Hence or otherwise factorise $P(x)$. 1
- b) Find the Cartesian equation of the curve given by the parametric equations: 2
- $$x = \cos \theta + \sin \theta$$
- $$y = \cos \theta - \sin \theta$$
- c) Nine people are partaking in a competition. In a particular round, the group of nine needs to split themselves into two teams of four and a score keeper.
- i) In how many ways can this be done? 2
- ii) These nine competitors are to be seated around a circular table for the competition. In how many ways can they be seated if the teammates of each team must sit together? 2
- d) How many five-digit numbers can be formed using digits 0, 2, 4, 5, 7 if none of the digits are repeated and the numbers formed are even? 2

End of Examination

Solutions (4r-11 Ext-1) Assessment Task-2, 2022

Section-I Multiple Choice

1) $|x-5| \leq 2$

$$-2 \leq x-5 \leq 2$$

$$3 \leq x \leq 7$$

(C)

2) $f(x) = x^3 - 4$

$$y = x^3 - 4$$

Swap x with y

$$x = y^3 - 4$$

$$y^3 = x + 4$$

$$y = \sqrt[3]{x+4}$$

$$f^{-1}(x) = \sqrt[3]{x+4}$$

(C)

3) $\left\lfloor \frac{22}{5} \right\rfloor = 5$

$$\text{or } 5 \times 4 + 2 = 22$$

$\therefore$ At least 1 box contains more than 4.

(B)

4) $6 \times 5 \times 4 = 120$

(C)

5) $P(x) = 2x^3 + ax - 3$

$$P(-1) = -1 \quad P(1) = ?$$

$$-2 - a - 3 = -1$$

$$a = -4$$

$$P(1) = 2 - 4 - 3 = -5$$

(B)

6) $P(x) = x^3 - x^2 - 5x - 3$

$$P'(x) = 3x^2 - 2x - 5$$

$$\text{Solve } P'(x) = 0$$

$$3x^2 - 2x - 5 = 0$$

$$x = -1, 5/3$$

$$\text{Since } P(-1) = 0$$

$$\therefore x = -1$$

(B)

7) (D)

8) $\frac{9!}{3! 2!}$

(C)

9) $(-\infty, 0) \cup \left[\frac{1}{5}, \infty\right)$

(A)

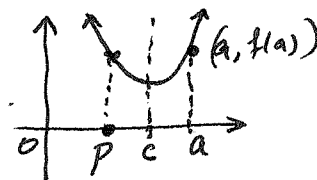
10) Since $x=c$ is the axis of symmetry of the parabola.

$$\therefore f(a) = f(2c-a)$$

$$f'(f(a)) = f'(f(2c-a))$$

$$= 2c - a$$

(B)



$$\text{At } P = c - (a - c) = 2c - a$$

Question-11a

$$\frac{2}{x-1} > 3, x \neq 1$$

Multiply both sides by $(x-1)^2$

$$2(x-1) > 3(x-1)^2$$

$$3(x-1)^2 - 2(x-1) < 0$$

$$(x-1)(3x-3-2) < 0$$

$$(x-1)(3x-5) < 0$$

$$1 < x < \frac{5}{3}$$



★ Can be done by using different methods.

11(b)

i) ${}^{12}C_5$ or 792 (1 mark)

ii) 2 women = ${}^7C_2 {}^5C_3 = 210$

1 woman = ${}^7C_1 {}^5C_4 = 35$

0 women = ${}^5C_5 = 1$

$$P(\text{at most 2 women}) = \frac{210 + 35 + 1}{792}$$

$$= \frac{246}{792} \text{ or } \frac{41}{132}$$

11(c)

i) $y = \frac{3}{x-2} + 1$

Swap x & y .

$$x = \frac{3}{y-2} + 1$$

$$\frac{3}{y-2} = x-1$$

$$y-2 = \frac{3}{x-1}$$

$$y = 2 + \frac{3}{x-1} \text{ or } f^{-1}(x) = 2 + \frac{3}{x-1}$$

3 marks correct solution & graph

2 marks for correct solution

1 mark for significant progress towards the sol.

2 marks correct ~~set~~ ^{answer}.
1 mark for significant progress towards the answer.

2 marks for correct inverse function.

1 mark for correct process to find inverse function

11(c)

$$\begin{aligned}\text{ii) } f^{-1}(f(x)) &= f^{-1}\left(\frac{3}{x-2} + 1\right) \\ &= \frac{3}{\frac{3}{x-2} + 1 - 1} + 2 \\ &= x - 2 + 2 \\ &= x\end{aligned}$$

(1 mark correct solution.
or correct process)

Hence shown.

11(d)

$$P(x) = 2x^3 - 10x^2 - 40x + 175$$

$$\text{i) } \alpha + \beta + \gamma = -\frac{b}{a} = \frac{10}{2} = 5$$

(1 mark)

$$\begin{aligned}\text{ii) } \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} \\ &= \frac{-40/2}{-175/2} \\ &= \frac{8}{35}\end{aligned}$$

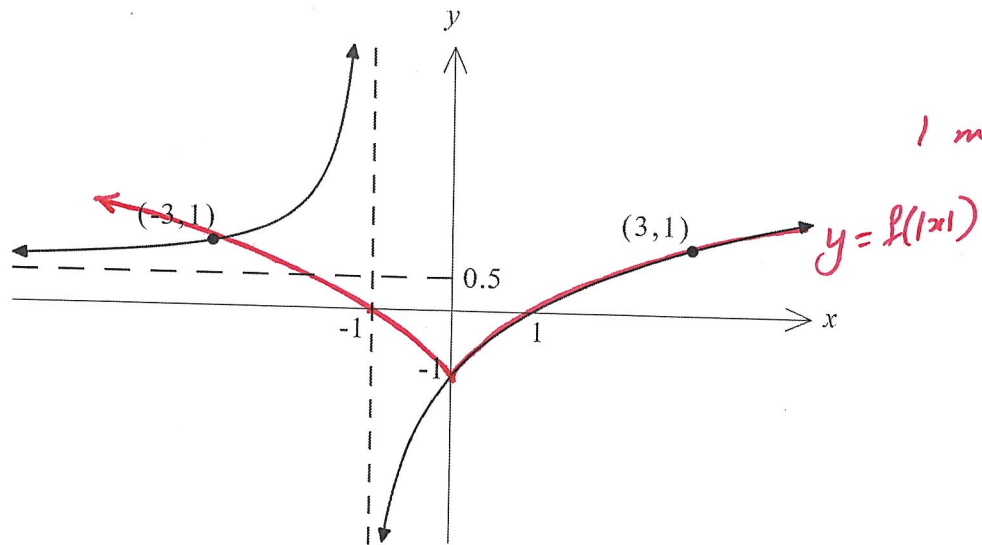
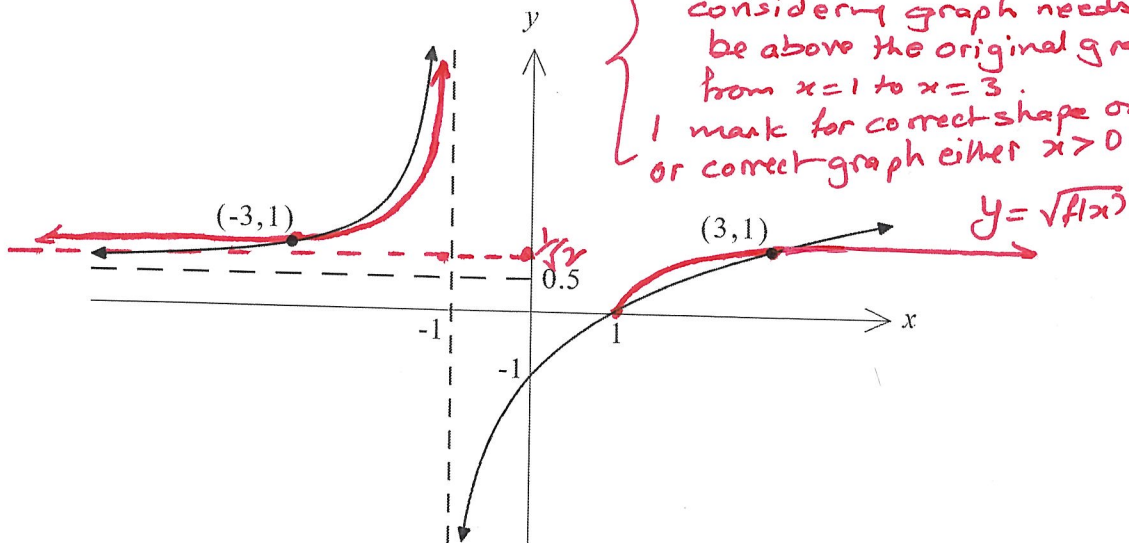
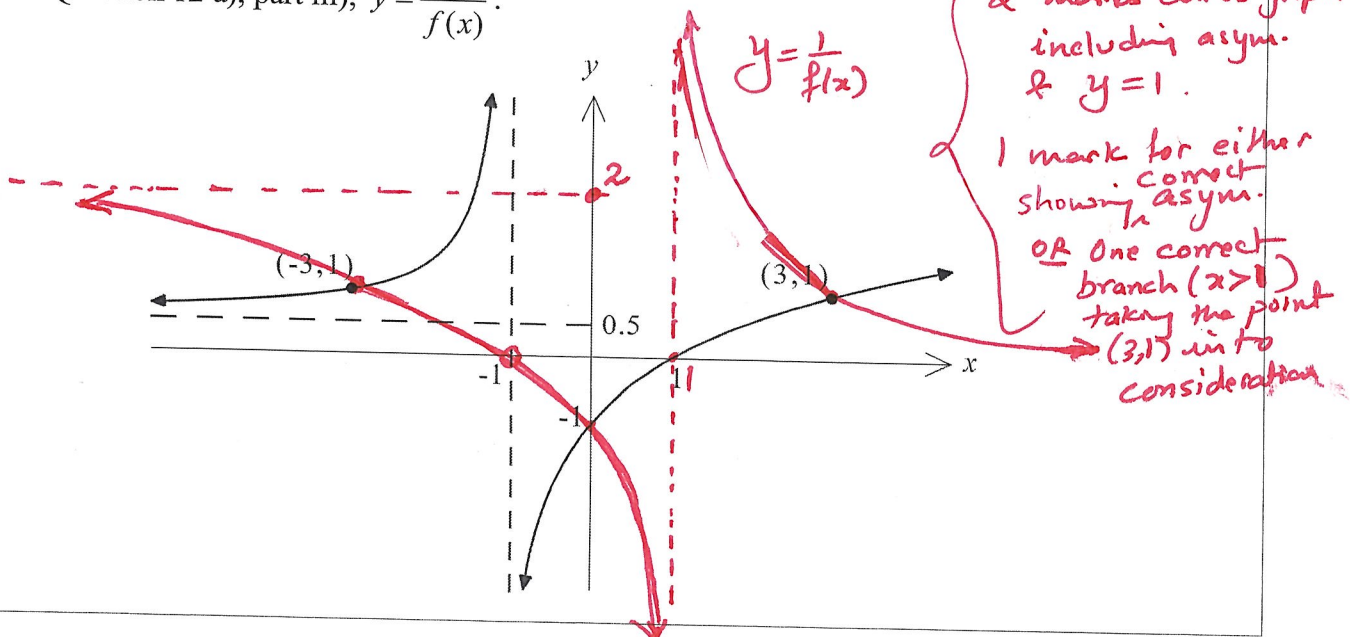
(2 marks correct answer
1 mark to simplify
 $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$)

$$\begin{aligned}\text{iii) } \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \\ &= (5)^2 - 2(-20) \\ &= 25 + 40 \\ &= 65\end{aligned}$$

(2 marks correct answer
1 mark correct process
towards the answer.
Correct ~~of~~
OR Expansion of $\alpha^2 + \beta^2 + \gamma^2$)

Question 12

Name:

Diagram for Question 12 a), part i), $y = f(|x|)$:Diagram for Question 12 a), part ii), $y = \sqrt{f(x)}$:Diagram for Question 12 a), part iii), $y = \frac{1}{f(x)}$:

Question 12(b)

12b i) 5 cubes (1 mark)
Worst case is r b r b r
 $r = \text{red}, b = \text{blue}$.

ii) 13 cubes. (1 mark)
Worst case is $r, r, \dots, r$, b b b
10 red

12c) $\frac{1}{x(x+2)} \leq -1, x \neq 0, -2$

$$\frac{1}{x(x+2)} + 1 \leq 0$$

$$\frac{1 + x(x+2)}{x(x+2)} \leq 0$$

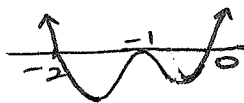
$$\frac{x^2 + 2x + 1}{x(x+2)} \leq 0$$

$$\frac{(x+1)^2}{x(x+2)} \leq 0$$

Multiply both sides by $x^2(x+2)^2$

$$x(x+2)(x+1)^2 \leq 0$$

$\therefore$ Solution is $-2 < x < 0$



* It can be done by using different methods

12d) $P(3) = 4, P(-5) = 12$

$$P(x) = (x-3)(x+5)Q(x) + (ax+b)$$

Since $P(3) = 4$

$$\therefore 4 = 3a + b \quad \text{--- ①}$$

& $P(-5) = 12$

$$12 = -5a + b \quad \text{--- ②}$$

Solve ① & ② simultaneously

$$a = -1, b = 7$$

$$\therefore \text{Remainder} = -x + 7$$

3 marks for correct solution
2 marks for significant progress (sketches polynomial or identifies $-2 \leq x \leq 0$)
1 mark to obtain $x(x+2)(x+1)^2 \leq 0$ or identifies 0, -1, -2 as critical values

2 marks correct solution
1 mark for evaluating $P(3)$ or $P(-5)$
OR Getting two equations correctly

Question-13

13a) i) BEGINNING

3N's 2G's 2I's 1B, 1E

(1 mark for correct answer)

G's together

No. of ways. $\frac{8!}{2! 3!} = 3360$

$\swarrow$ $\nwarrow$
 2 identical I's 3 identical N's

ii) Total Arrangements = $\frac{9!}{2! 2! 3!} = 15120$

$\swarrow$ $\uparrow$ $\nwarrow$
 2G's 2I's 3N's

2 marks for correct solution.
1 mark for taking 3N's case but not 2N's

3N's together = $\frac{7!}{2! 2!} = 1260$

2N's together. $\underbrace{NN} - \overset{\text{Third}}{\underset{\text{N can be placed at 6 places}}{\times \times \times \times \times \times}} \dots 6 \text{ places.}$

||ly If two N's are on other end. $\dots 6 \text{ places.}$

$\begin{array}{c}
 \underbrace{NN} - \times \times \times \times \times \\
 \times - NN - \times \times \times \times \\
 - - - NN - - - - \\
 \vdots \\
 NN -
 \end{array}
 \left. \begin{array}{l}
 \text{Third N can be placed at 5 places} \\
 \text{" " " " " " " 5 " }
 \end{array} \right\} \text{Total ways are 6}$

$\therefore \text{Total} = 6 + 6 + 5 \times 6 = 42$

No. of ways = $42 \times \frac{6!}{2! \times 2!} = 7560$

Using Complementary N's are NOT adjacent i's

Total - 3N's together - 2N's together

$15120 - 1260 - 7560 = 6300.$

Method-2.

$\times B \times E \times G \times I \times I \times G \times$

N can be placed at 7 different places so that they are not adjacent.

$\frac{{}^7P_3}{\rightarrow 3!} \times \frac{6!}{2! \times 2!} = 6300$

$\swarrow$ $\nwarrow$
 3 Identical N's 2G's 2I's

2 marks for correct sol.
1 mark if they just write answer without explanation

Question 13

13b) $f(x) = x^2 + 4x + 1$

i) The domain has to be restricted as the given function is not one to one or does not pass the horizontal line test. (1 mark)

ii) Largest domain containing $x=0$ is $x \geq -2$ (1 mark)

iii) $y = x^2 + 4x + 1, x \geq -2$

Swap x and y

$$x = y^2 + 4y + 1$$

$$x = y^2 + 4y + 4 - 3$$

$$x + 3 = (y + 2)^2$$

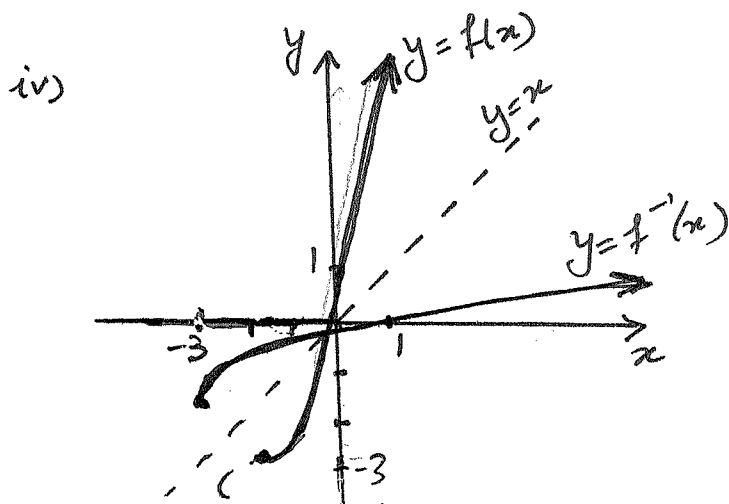
$$y + 2 = \pm \sqrt{x + 3}$$

$$y = -2 + \sqrt{x + 3} \quad (\because y \geq -2)$$

$$f^{-1}(x) = -2 + \sqrt{x + 3}$$

Hence shown.

{ 2 marks for correct process
1 mark if $y \geq -2$ is not taken into consideration



{ 2 marks for each graph drawn correctly with all the important points clearly mentioned.
1 mark for either $f(x)$ & $f^{-1}(x)$ drawn correctly label points as well.

Question-13

13(c). $x^3 + px^2 - qx + 3 = 0$

Let roots are $\alpha, \frac{1}{\alpha}, \beta$

Product of roots $= -\frac{d}{a} = -3$

$$\therefore \alpha \times \frac{1}{\alpha} \times \beta = -3$$

$$\beta = -3$$

$\therefore$ Third root is -3 .

(1 mark correct solution)

ii) Since -3 is a root

$$\therefore (-3)^3 + p(-3)^2 - q(-3) + 3 = 0$$

$$-27 + 9p + 3q + 3 = 0$$

$$9p + 3q = 24$$

$$3p + q = 8$$

$$p = \frac{8-q}{3}$$

2 marks correct solution
1 mark for significant progress towards answer
OR correct sum of roots & sum of two at a time including $\beta = -3$

OR. Sum of roots $= -\frac{b}{a} = -p$

$$\alpha + \frac{1}{\alpha} - 3 = -p$$

$$\alpha + \frac{1}{\alpha} = 3 - p \quad \text{--- (1)}$$

$\therefore$ Sum of product of two at a time $= \frac{c}{a} = -q$

$$1 - 3\alpha - \frac{3}{\alpha} = -q$$

$$1 - 3\left(\alpha + \frac{1}{\alpha}\right) = -q$$

$$1 - 3(3 - p) = -q$$

$$1 - 9 + 3p = -q$$

$$-8 + 3p = -q$$

$$p = \frac{8-q}{3}$$

(from ①)

Question-14

14a $P(x) = 4x^3 - 13x^2 + 4x + 12$

i) $P'(x) = 12x^2 - 26x + 4$

Solve $P'(x) = 0$

$$12x^2 - 26x + 4 = 0$$

$$(6x-1)(x-2) = 0$$

$$x = \frac{1}{6} \text{ or } 2$$

{ 2 marks for correct solution.
1 mark for solving $P'(x) = 0$

Test for $x = \frac{1}{6}$, $P(\frac{1}{6}) \neq 0$

Test for $x = 2$, $P(2) = 0$

$$\therefore 4(2)^3 - 13(2)^2 + 4(2) + 12 = 0$$

$\therefore x = 2$ is double root of $P(x)$ as $P(2) = 0$ & $P'(2) = 0$

$$\alpha = 2$$

ii) $P(x) = (x-2)^2(ax+b)$

(1 mark)

By comparing coeff. of x^3 & constant

$$a = 4, b = 3$$

$$\therefore P(x) = (x-2)^2(4x+3)$$

OR Can be done by using long division.

b) $x = \cos \theta + \sin \theta$ ——— ①

$y = \cos \theta - \sin \theta$ ——— ②

{ 2 marks finding correct Cartesian equation
1 mark towards significant progress to eliminate θ .

$$\textcircled{1}^2 + \textcircled{2}^2$$

$$x^2 + y^2 = \cos^2 \theta + \sin^2 \theta + 2\sin \theta \cos \theta + \cos^2 \theta + \sin^2 \theta - 2\sin \theta \cos \theta$$
$$= 1 + 1$$

$$\therefore x^2 + y^2 = 2.$$

Question-14

14(c)i) $\frac{{}^9C_4 \times {}^5C_4 \times 1}{2} = 315$

2 $\leftarrow$ for two identical groups.

(2 marks correct sol.
1 mark for ${}^9C_4 \times {}^5C_4$)

ii) $2! \times 4! \times 4! = 1152$

arranging
the groups.

arranging teammates
within the groups.

(2 marks for correct sol.
1 mark $4! \times 4!$
OR $2 \times 5! \times 5!$)

14(d). Five-digit Number $\{0, 2, 4, 5, 7\}$

Number starting with even numbers 2, 4, (0 can't take that place)

$$\underbrace{2}_{\substack{\uparrow \\ \text{Two options}}} \times \underbrace{3 \times 2 \times 1}_{3!} \times \underbrace{2}_{\substack{\uparrow \\ \text{Two options}}} = 24$$

Number starting with odd numbers 5, 7

$$\underbrace{2}_{\substack{\uparrow \\ \text{Two options}}} \times \underbrace{3 \times 2 \times 1}_{3!} \times \underbrace{3}_{\substack{\uparrow \\ \text{Three options}}} = 36$$

$$\text{Total no. of ways} = 24 + 36 = 60$$

2 marks correct solution
1 mark for significant progress/clear explanation towards the answer.

The End.